

HDOC DAY-29 ASSIGNMENT

Menu-Driven Array Operations Using Switch-Case

1. Problem Understanding

The task is to write a menu-driven C program using switch-case to perform operations on one-dimensional integer arrays. The program provides the following options:

- Count even, odd, positive, negative, and zero elements in an array.
- Search for an element in an array using linear search.
- Merge two arrays.
- Exit the program.

2. Standard Test Cases

Test Case	Choice	All Inputs	Expected Output	Status
TC1	1	n=6; array={2,-3,0,8,-5,7}	Even=2, Odd=3, Positive=3, Negative=2, Zero=1	Pass
TC2	2	n=5; array={10,20,30,40,50}; key=30	Element found at position 3	Pass
TC3	2	n=4; array={5,9,12,18}; key=7	Element not found	Pass
TC4	3	n1=3; A={1,2,3}; n2=3; B={4,5,6}	Merged array={1,2,3,4,5,6}	Pass
TC5	4	No array input	Program exits	Pass

3. Algorithm

A. Common Menu Logic

1. Start.
2. Display the menu options.
3. Read the user's choice.
4. Use switch-case to select the required array operation.
5. For choices 1 to 3, read the required array size(s) and elements inside the selected operation.
6. Execute the selected operation and display the result.
7. For choice 4, terminate the program.
8. For any other choice, display "Invalid choice".
9. Stop.

B. Count Even, Odd, Positive, Negative, and Zero Elements

1. Read the number of elements n.
2. Read n integer elements into the array.
3. Initialize even, odd, positive, negative, and zero counters to 0.
4. Traverse the array from index 0 to n - 1.
5. If an element is divisible by 2, increment even; otherwise increment odd.

6. If an element is greater than 0, increment positive; if less than 0, increment negative; otherwise increment zero.
7. Display all five counts.
8. Stop.

C. Linear Search

1. Read the number of elements n.
2. Read n integer elements into the array.
3. Read the element to be searched as key.
4. Set position = -1.
5. Traverse the array from index 0 to n - 1.
6. If array[i] equals key, store i + 1 in position and stop searching.
7. If position remains -1, display "Element not found"; otherwise display its position.
8. Stop.

D. Merge Two Arrays

1. Read the size n1 of the first array and its elements.
2. Read the size n2 of the second array and its elements.
3. Copy all elements of the first array into the merged array.
4. Copy all elements of the second array after the first array elements.
5. Display the merged array containing n1 + n2 elements.
6. Stop.

4. C Program

```
#include <stdio.h>

void countElements()
{
    int n, i, a[100];

    int even = 0, odd = 0, positive = 0, negative = 0, zero = 0;

    printf("Enter number of elements: ");

    scanf("%d", &n);

    printf("Enter array elements: ");

    for (i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    for (i = 0; i < n; i++)
    {
        if (a[i] % 2 == 0)
```

```

    {
        even++;
    }
    else
    {
        odd++;
    }
    if (a[i] > 0)
    {
        positive++;
    }
    else if (a[i] < 0)
    {
        negative++;
    }
    else
    {
        zero++;
    }
}
printf("Even = %d\n", even);
printf("Odd = %d\n", odd);
printf("Positive = %d\n", positive);
printf("Negative = %d\n", negative);
printf("Zero = %d\n", zero);
}

void linearSearch()
{
    int n, i, key, position = -1, a[100];
    printf("Enter number of elements: ");
    scanf("%d", &n);

```

```

printf("Enter array elements: ");
for (i = 0; i < n; i++)
{
    scanf("%d", &a[i]);
}
printf("Enter element to search: ");
scanf("%d", &key);
for (i = 0; i < n; i++)
{
    if (a[i] == key)
    {
        position = i + 1;
        break;
    }
}
if (position == -1)
{

    printf("Element not found\n");
}
else
{
    printf("Element found at position %d\n", position);
}
}

void mergeArrays()
{
    int n1, n2, i, a[100], b[100], merged[200];
    printf("Enter size of first array: ");
    scanf("%d", &n1);
    printf("Enter first array elements: ");

```

```

for (i = 0; i < n1; i++)
{
    scanf("%d", &a[i]);
    merged[i] = a[i];
}

printf("Enter size of second array: ");
scanf("%d", &n2);
printf("Enter second array elements: ");
for (i = 0; i < n2; i++)
{
    scanf("%d", &b[i]);
    merged[n1 + i] = b[i];
}

printf("Merged array: ");
for (i = 0; i < n1 + n2; i++)
{
    printf("%d ", merged[i]);
}

printf("\n");
}

int main()
{
    int choice;
    do
    {
        printf("\nMENU\n");
        printf("1. Count even, odd, positive, negative and zero elements\n");
        printf("2. Linear search\n");
        printf("3. Merge two arrays\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
    }

```

```
scanf("%d", &choice);  
switch (choice)  
{  
    case 1:  
        countElements();  
        break;  
    case 2:  
        linearSearch();  
        break;  
    case 3:  
        mergeArrays();  
        break;  
    case 4:  
        printf("Program exited\n");  
        break;  
    default:  
        printf("Invalid choice\n");  
}  
}  
while (choice != 4);  
return 0;  
}
```

5. Compilation and Execution Evidence

A screenshot of a Kali Linux desktop environment. The top panel shows various application icons and system status indicators like time (12:43) and battery level. A terminal window titled "kali@kali: ~/Desktop" is open, displaying the output of running a script called ./question. The background features a faint anime-style illustration of a character with glasses.

```
kali@kali:~/Desktop$ ./question\nMENU\n1. Count even, odd, positive, negative and zero elements\n2. Linear search\n3. Merge two arrays\n4. Exit\nEnter your choice: 1\nEnter number of elements: 6\nEnter array elements: 2 -3 0 8 -5 7\nEven = 3\nOdd = 3\nPositive = 3\nNegative = 2\nZero = 1\n\nMENU\n1. Count even, odd, positive, negative and zero elements\n2. Linear search\n3. Merge two arrays\n4. Exit\nEnter your choice: 2\nEnter number of elements: 5\nEnter array elements: 10 20 30 40 50\nEnter element to search: 30\nElement found at position 3\n\nMENU\n1. Count even, odd, positive, negative and zero elements\n2. Linear search\n3. Merge two arrays\n4. Exit\nEnter your choice: ^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B^[[B
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
Enter array elements: 10 20 30 40 50
Enter element to search: 30
Element found at position 3

MENU
1. Count even, odd, positive, negative and zero elements
2. Linear search
3. Merge two arrays
4. Exit
Enter your choice: 2
2
Enter number of elements: 4
Enter array elements: 5 9 12 18
Enter element to search: 7
Element not found

MENU
1. Count even, odd, positive, negative and zero elements
2. Linear search
3. Merge two arrays
4. Exit
Enter your choice: 3
Enter size of first array: 3
Enter first array elements: 1 2 3
Enter size of second array: 3
Enter second array elements: 4 5 6
Merged array: 1 2 3 4 5 6

MENU
1. Count even, odd, positive, negative and zero elements
2. Linear search
3. Merge two arrays
4. Exit
Enter your choice: 4
Program exited

(kali@kali)-[~/Desktop]
$
```

6. Verification of All Test Cases

TC	Choice	All Inputs	Expected Result	Actual Result	Status
TC1	1	n=6; {2,-3,0,8,-5,7}	E=3, O=3, P=3, N=2, Z=1	E=3, O=3, P=3, N=2, Z=1	Pass
TC2	2	n=5; {10,20,30,40,50}; key=30	Found at position 3	Found at position 3	Pass
TC3	2	n=4; {5,9,12,18}; key=7	Element not found	Element not found	Pass
TC4	3	n1=3; {1,2,3}; n2=3; {4,5,6}	{1,2,3,4,5,6}	{1,2,3,4,5,6}	Pass
TC5	4	No array input	Program exits	Program exits	Pass